Consensus-Driven Event-Based Graph Information Bottleneck for Integrated Communication and Control

Ziqiong Wang, Xiaoxue Yu, Rongpeng Li, *Senior Member, IEEE*, and Zhifeng Zhao

***Abstract*—The integration of communication and control presents a critical challenge for Uncrewed Aerial Vehicle (UAV) swarms operating in the formation control scenario. This co-design problem is typically formulated within the Multi-Agent Reinforcement Learning (MARL) framework. However, partial observability limitations will impair collaboration effectiveness, and a potential solution is to establish consensus through well-calibrated latent variables obtained from neighboring agents. Nevertheless, the rigid transmission of less informative content can still result in redundant information exchanges. Therefore, we propose a Consensus-Driven Event-Based Graph Information Bottleneck (CDE-GIB) method, which integrates the communication graph and information flow through a GIB regularizer to extract more concise message representations while avoiding the high computational complexity of inner-loop operations. Meanwhile, we investigate the potential impact of consensus errors induced by noise arising from multi-agent interactions. To further minimize the communication volume required for establishing consensus during interactions, we also develop a variable-threshold event-triggering mechanism. By simultaneously considering historical data and current observations, this mechanism capably evaluates the importance of information to determine whether an event should be triggered. Experimental results demonstrate that our proposed method outperforms existing state-of-the-art methods in terms of efficiency, adaptability, and noise robustness.**

***Index Terms*—Communication and control co-design, event trigger, graph information bottleneck optimization, consensus-oriented, multi-agent reinforcement learning.**

I. Introduction

NOWADAYS, Uncrewed Aerial Vehicles (UAVs) have gained significant attention in both military and civilian realms, demonstrating great potential for complex tasks such as formation control [1] due to their agility and compact size. Nevertheless, the performance of UAV swarms is critically challenged by the fundamental issue of integrating communication and control [2]. Typically, such a co-design is contingent on Multi-Agent Reinforcement Learning (MARL) [3], but suffers from partial observability [4], as agents can only access limited or incomplete information about the environment and other agents' states. Therefore, in order to make informed collaborative decisions, it necessitates the information sharing and aggregation among agents [5]. Although full-mesh, raw data transmission [6] could theoretically resolve this problem, the overwhelming information exchange often makes it practically infeasible, especially under bandwidth-limited scenarios. Until recently, algorithms have started to emerge [7], establishing consensus over well-calibrated latent variables from neighbors only. Compared to direct data transmission, the consensus-based algorithm boosts scalable decentralized execution. Nevertheless, the insufficiency of communication and control co-design, such as the blunt transmission of less informative messages [8], leaves significant redundancy in exchanged messages. Correspondingly, the inevitable difficulties posed by limited bandwidth [9] and noisy channels [10] still call for a more efficient algorithmic framework.

As a remedy, theoretically guided refinement and compression contribute to squeezing the amount of exchanged information for consensus inference. In that regard, the integration of Information Bottleneck (IB) emerges as a highly promising direction to improve overall communication efficiency. In the context of general representation learning, the IB principle [11] emphasizes that the optimal representation should contain sufficient and minimal information that is beneficial for ultimate tasks. Concurrently, Task-Oriented Communication Framework (TOCF) [12] introduces IB theory into the Multi-Agent Communication Reinforcement Learning (MACRL) scenarios, enabling efficient message compression in communication. However, such IB-based representation learning methods typically require input data to meet independent and identically distributed (i.i.d.) conditions, which do not always hold in the context of multi-agent communication. Meanwhile, Ref. [12] overlooks that both graph structure and agent features carry important information in MACRL, whereas Graph Information Bottleneck (GIB) theory [13] introduces a local-dependence assumption and provides a paradigm that regularizes the topological as well as the feature attributes, offering significant advantages. While Multi-Agent communication mechanism with Graph Information bottleneck optimization (MAGI) [14] extends the GIB principle to MACRL methods and derives more effective

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Ziqiong Wang, Xiaoxue Yu, and Rongpeng Li are with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou 310058, China (e-mail: wangziqiong@zju.edu.cn; sdwhyxx@zju.edu.cn; lirongpeng@zju.edu.cn).

Zhifeng Zhao is with the Zhejiang Lab, Hangzhou 311121, China, and also with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou 310058, China (e-mail: zhaozf@zhejianglab.org).

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and concise message representations, its reliance on Graph Neural Networks (GNN) [15] for inter-agent communication introduces critical vulnerabilities [16]. Recent study [17] has revealed that even minor disturbances in node features or edge weights can catastrophically degrade GNN's message-passing efficacy. Furthermore, the separate optimization of the communication graph and information flow in MAGI poses structural limitations, while the underlying full-mesh communication with all agents within a certain range inevitably adds significantly to the communication volume, necessitating the design of a more efficient and robust information compression approach. In this regard, event-triggered mechanisms [18], [19] can contribute to reducing redundant exchanged information by appropriately tuning the frequency of communications. However, these methods often determine the triggering moment solely based on current observations, but neglect the influence of historical information. In contrast, Event-Triggered Communication Network (ETCNet) [20] enables agents to make more informed decisions by leveraging both immediate and historical data. Nevertheless, it fails to offer a detailed explanation regarding the criteria for setting the threshold, while Ref. [21] relies on fixed thresholds, potentially encountering accumulative errors due to stale updates. Furthermore, the effective incorporation of event-triggered communications into consensus inference still awaits for comprehensive investigation.

In this paper, we propose the Consensus-Driven Event-based GIB (CDE-GIB) algorithm for the co-design of communication and control in the formation task. Unlike Vector Quantization-Graph Information Bottleneck (VQ-GIB) [22] that applies the GIB principle to static subgraph classification tasks, our method extends the idea to dynamic multi-agent coordination by introducing distance-based relational coding as the GIB input. In addition, compared to the existing approach that separately compresses the communication graph and data flow while treating the communication pathway as an implicit closed-box component [14], we develop a more communication-efficient GIB regularizer and integrate a Variable-Threshold Event-Triggering Mechanism (VT-ETM). Specifically, the VT-ETM takes account of both historical data and current observations and determines whether to trigger a communication transmission event from the perspective of information importance. In contrast to MAGI, where all neighboring messages are first exchanged before attention-based filtering, our method proactively determines the communication necessity prior to messages transmission. Furthermore, we refine the model to explicitly incorporate the accumulative errors, arising from multi-agent interactions, for consensus inference. In comparison to existing works in the literature, the contribution of this paper can be summarized as follows.

- We propose the CDE-GIB framework for formation control with efficient communications, which novelly combines event-triggering and GIB, to ameliorate the message inefficiency of consensus inference in decentralized MARL. By considering cumulative error-induced perturbations, we enhance both the efficiency and robustness of downstream control tasks.
- We develop a VT-ETM that dynamically evaluates the information's importance towards inferring the consensus across multiple agents, distinguishing it significantly from existing MARL-empowered solutions [18], [19], [20]. We design a devised GIB regularizer that fuses the communication graph and information flow to obtain more concise message representations.
- We validate the universal effectiveness of our framework through extensive simulations in the multi-agent particle environment [23], and demonstrate its superiority over state-of-the-art approaches like Consensus-oriented Multi-Agent Communication (ConsMAC) [24], Targeted Multi-Agent Communication (TarMAC) [5] and Multi-Agent communication via Self-supervised Information Aggregation (MASIA) [6].

The remainder of the paper is organized as follows. Section II succinctly reviews the related works. Section III briefly introduces the system model and formulates the problem. Section IV presents the overview of our proposed CDE-GIB framework. In Section V, we elaborate on the experimental results and discussions. Finally, Section VI concludes the paper.

II. Related Work

A. Graph Information Bottleneck in MACRL

Representative MACRL methods typically employ attention-based communication [5] or opponent modeling [27] to alleviate partial observability. ConsMAC [24] further improves decentralized cooperation by guiding agents to integrate local observations and communication messages for consistent global-state estimation. However, existing MACRL communication mechanisms often lack communication and control co-design, which may result in inefficient message exchange. To address this limitation, IB [11] provides a theoretically guided approach to refine and compress exchanged information. While TOCF [12] compresses information flow, it overlooks the graph-structured interdependencies essential for agent coordination. To incorporate structural awareness, Ref. [26] introduces a subgraph generator for edge pruning and node filtration, preserving a compressed yet information-rich substructure. However, this approach relies solely on current information for node retention decisions, disregarding the potential contributions of historical data. Moreover, several recent GIB-based methods focus primarily on static graph processing or classification tasks. For instance, although the framework in [22] demonstrates notable robustness against channel noise, it is not readily applicable to dynamic multi-agent collaborative settings. In contrast, this paper proposes a novel GIB method that extends conventional IB frameworks to simultaneously incorporate graph structural awareness and information flow. Furthermore, we seamlessly integrate an event-triggered mechanism capable of evaluating the semantic importance of information.

B. Event-Triggered Implementation Based on Consensus Inference

To enhance communication efficiency in MAS, event-triggered mechanisms have emerged to identify and select the most relevant agents for information exchange. Individualized Controlled Continuous Communication Model (IC3Net) [28] restricts each agent to a binary broadcast decision, thereby

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TABLE I
SUMMARY AND COMPARISON OF RELATED SURVEY

References	Reliable Consensus Establishment	Information Compression	Event-Triggered Mechanism	Brief Description
Y. Xiang, *et al.* [24]	●	○	○	Leaving significant redundancy in exchanged messages.
G. He, *et al.* [12]	○	◑	○	Only focusing on compressing observations into feature representations.
S. Ding, *et al.* [14]	○	◑	○	Complex computations from separate optimization of the communication graph and information flow.
C. Liu *et al.* [25]	●	○	●	Exclusive reliance on reducing the communication frequency.
L. Zhong, *et al.* [26]	○	●	◑	Using solely current local observations for communication agent selection.
This paper	●	●	●	Combining communication-efficient VT-ETM with representation-optimized GIB into decentralized MARL collaboration under establishing consensus.

Notations: ● *indicates fully included;* ◑ *means partially included;* ○ *denotes not included.*

ignoring heterogeneity in receivers' information needs. Likewise, Gated-Actor-Critic Message Learner (Gated-ACML) [29] also fails to address this important aspect and determines the triggering moment solely based on current observations. While QD-learning considers both historical and immediate information, it lacks a detailed explicit formulation for setting the communication threshold [30]. In contrast, our proposed VT-ETM synthesizes the aforementioned advancements while addressing their limitations by leveraging both historical data and current observations to assess information criticality. Our scheme not only determines when to trigger communication but also selects with whom to communicate, all governed by a variable threshold strategy that evolves over time to ensure adaptability and efficiency.

In a nutshell, we summarize the key differences between our algorithm and relevant literature in Table I.

III. PRELIMINARIES AND SYSTEM MODEL

Beforehand, we summarize the main notations in Table II.

TABLE II
MAJOR NOTATIONS USED IN THE PAPER

Notation	Definition
$\mathbf{s}^{(t)}, \mathbf{o}_g^{(t)}$	Global state and observation at time step t.
$\mathbf{z}_i^{(t)}, \mathbf{u}_i^{(t)}$	Local state, individual action of agent i at time step t.
$\mathbf{o}_i^{(t)}, \mathbf{m}_i^{(t)}$	Local observation, exchanged message of agent i at time step t.
$\mathbf{v}_i^{(t)}, \mathbf{p}_i^{(t)}$	Velocity and position that constitute the local observation of agent i at time step t.
$\mathbf{E}_{m_i}^{(t)}$	Aggregated message embedding of agent i at time step t.
$\mathbf{h}^{(t)}, \hat{\mathbf{e}}^{(t)}$	Global latent representation and estimated global observation at time step t.
$\mathbf{E}_{o_i}^{(t)}, \varpi_i^{(t)}$	Embedding vector and communication information weight of agent i at time step t.
$\xi_i^{(t)}$	Neighbors within the observation range of agent i at time step t.
$\beta_i^{(t)}$	Importance ratio of MAPPO of agent i at time step t.
α	Coefficient of policy entropy.
π_{θ_i}, θ_i	Target policy and its parameter of agent i.
$\pi_{\theta_{i,\text{old}}}, \theta_{i,\text{old}}$	Behavior policy and its parameter of agent i.
V_ϕ, ϕ	Current value function and its parameter.
$V_{\phi_{\text{old}}}, \phi_{\text{old}}$	Previous value function and its parameter.
Θ, Ψ	Parameters of the policy.
$\mathcal{T}_i$	Sequence of event-triggering times of agent i.
N	Number of UAVs.
$\mathcal{B}$	Replay memory.
B_n	Training batch size.
$G_{\text{threshold}}$	Threshold of VT-ETM.
$\mathcal{A}_i^{(t)}$	Event triggered behavior of agent i.

A. Theoretical Background

1) Consensus: Consensus denotes the process through which a team of agents reaches agreement on a shared value or state variable through local communication and distributed computation. In UAV swarm formation control, consensus algorithms [5], [6], [24] enable agents to estimate the global observation $\mathbf{o}_g^{(t)}$ at time t using the local observation $\mathbf{o}_i^{(t)} = \{(\mathbf{p}_j^{(t)}, \mathbf{v}_j^{(t)}) \mid \forall j \in \xi_i^{(t)}\}$ composed of the positions $\mathbf{p}_j^{(t)}$ and velocities $\mathbf{v}_j^{(t)}$ of agent i and its neighbors and exchanged messages $\{\mathbf{m}_j^{(t)} \mid \forall j \in \xi_i^{(t)}\}$, where $\xi_i^{(t)}$ comprises agent i and its neighbors. This distributed coordination mechanism is essential for achieving and maintaining cohesive group behavior under communication constraints.

2) Event-Triggered Mechanism: Event-triggered Mechanism reduces communication overhead by transmitting messages only when specific conditions are met, rather than employing periodic communication. A communication event is triggered when a triggering function $\kappa(\cdot)$ exceeds a predetermined threshold $G_{\text{threshold}}$.

3) Information Bottleneck: The IB principle provides an information-theoretic framework for learning compressed representations that preserve task-relevant information while discarding irrelevant details. Given input data X and target output Y, the IB objective is to maximize the mutual information between the learned representation Z and Y, while minimizing the mutual information between Z and X

$$\max_{p_{\theta_e}(Z|X)} \mathcal{I}(Y;Z) - \beta \mathcal{I}(Z;X), \tag{1}$$

where β controls the trade-off between compression and preservation of relevant information, and θ_e denotes the learnable parameters of the encoder $p_{\theta_e}(Z|X)$ that maps the input X to the latent representation Z.

4) Graph Neural Network: GNNs are a class of deep learning architectures that operate on graph-structured data. A graph

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$\mathcal{G} = (\mathcal{N}, \mathcal{E}, \mathcal{H})$ consists of nodes $\mathcal{N}$, edges $\mathcal{E}$ and nodes' attributes $\mathcal{H}$. By performing iterative message passing and aggregating operations, GNNs enable each node in a graph to incorporate and refine information from its topological neighbors. GNNs facilitate distributed reasoning and collaborative behavior by allowing agents to efficiently share and update information across the communication network.

B. System Model

1) UAV Dynamic Model: The dynamic behavior of each UAV agent i is governed by a double-integrator model, a standard formulation for capturing the motion of aerial vehicles. The dynamic properties of UAV i at time t are characterized by its position $\mathbf{p}_i^{(t)} = (p_{x_i}^{(t)}, p_{y_i}^{(t)}, p_{z_i}^{(t)})$ and velocity $\mathbf{v}_i^{(t)} = (v_{x_i}^{(t)}, v_{y_i}^{(t)}, v_{z_i}^{(t)})$ in a three-dimensional North-East-Down (NED) inertial frame. The position evolution is controlled by the acceleration $\mathbf{u}_i^{(t)} = (u_{x_i}^{(t)}, u_{y_i}^{(t)}, u_{z_i}^{(t)})$, yielding the following continuous-time dynamics,

$$\dot{\mathbf{p}}_i^{(t)} = \mathbf{v}_i^{(t)}, \tag{2}$$

$$\dot{\mathbf{v}}_i^{(t)} = \mathbf{u}_i^{(t)}. \tag{3}$$

Aligned with MARL-related mainstream UAVs studies for formation control task [31], [32], [33], we assume the UAV swarm operates at a fixed altitude, by constraining $p_{z_i}^{(t)} \equiv 10$ m and $u_{z_i}^{(t)} \equiv 0$.

2) Channel Model: We assume that the continuous-valued communication channel between transmitting and receiving agents is a Rician fading channel. The received signal $\hat{\mathbf{x}}$ is given by [34]

$$\hat{\mathbf{x}} = h \cdot \mathbf{x} + \epsilon, \tag{4}$$

where the transmitted signal $\mathbf{x} \in \mathbb{R}^K$ is impaired by the complex Rician channel gain $h = \sqrt{\frac{K_L}{K_L+1}}\, h_{\text{LOS}} + \sqrt{\frac{1}{K_L+1}}\, h_{\text{NLOS}}$, with the Rician K_L-factor denoting the LOS-to-NLOS power ratio, and $\epsilon \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$ models the complex Additive White Gaussian Noise (AWGN) [35], [36].

C. Task Model

We primarily consider a decentralized, communication-limited formation control task, where a set of UAVs navigates toward a destination Δ_p while achieving a pre-defined formation. Notably, the formation Δ_f^N is N-dependent, as illustrated in Fig. 1.

In order to accomplish the formation control task, we formulate the problem as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) [37], which is characterized by a tuple $\langle \mathcal{N}, \mathcal{S}, \mathcal{U}, \mathcal{P}, \mathcal{R}, \Omega, \Lambda, \gamma \rangle$. $\mathcal{N}$ represents the set of N active nodes. Due to the practical communication limitation, the heterogeneous connectivity across nodes can be characterized by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ – $A(i, j) = 1$ if and only if the Euclidean distance between $i \in \mathcal{N}$ and $j \in \mathcal{N}$ is less than the maximum observation range δ_{com}; and it nulls otherwise. Naturally, the MAS can be denoted as a graph $\mathcal{G} = (\mathcal{N}, \mathcal{E}, \mathcal{H})$, where $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ represents the edge set characterized by $\mathbf{A}$, and $\mathcal{H} = \{\mathbf{h}^{(t)} \mid t = 1, 2, \ldots, T_{\text{step}}\}$ contains the feature attributes of the nodes. $\mathcal{S}$ denotes the global state space of the problem and $\mathcal{U}$ is the homogeneous action space for the multi-agent system. The joint action $\mathbf{u}^{(t)} = \{\mathbf{u}_i^{(t)} \mid \forall i \in \mathcal{N}\}$ executed at the current state $\mathbf{s}^{(t)}$ makes the environment transit to the next state $\mathbf{s}^{(t+1)}$ according to the transition probability function $\mathcal{P}(\mathbf{s}^{(t+1)} \mid \mathbf{s}^{(t)}, \mathbf{u}^{(t)}) : \mathcal{S} \times \mathcal{U} \times \mathcal{S} \to [0, 1]$. Due to the limited capacity for perception of the complex environment, each agent i acquires a local observation $\mathbf{o}_i^{(t)} \in \Omega$ via the observation function $\Lambda(\mathbf{o}_i^{(t)} \mid \mathbf{s}_i^{(t)}, i) : \mathcal{S} \times \mathcal{N} \times \Omega \to [0, 1]$. All agents share a global reward function $\mathcal{R}(\mathbf{s}^{(t)}, \mathbf{u}^{(t)}) : \mathcal{S} \times \mathcal{U} \to \mathbb{R}$ and the overall objective is to maximize the total discounted reward $\mathbb{E}[\sum_t \gamma^t \mathcal{R}^{(t)}]$, where $\gamma \in [0, 1]$ means a discount factor. In alignment with the Dec-POMDP framework, we specify the elements as follows.

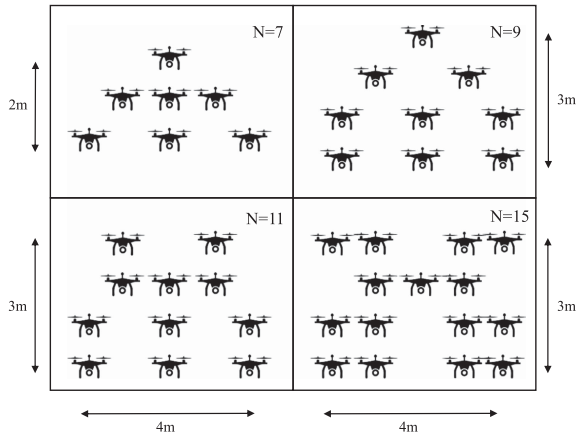


Fig. 1. An illustration of the pre-defined formations corresponding to different numbers of UAVs (i.e., $N \in \{7, 9, 11, 15\}$).

1) State: The state $\mathbf{s}^{(t)} \in \mathcal{S}$ can be task-dependent. For example, in an UAV scenario, an agent i can obtain direct observation $\mathbf{o}_i^{(t)} = \{(\mathbf{p}_j^{(t)}, \mathbf{v}_j^{(t)}) \mid \forall j \in \xi_i^{(t)}\}$ composed of the positions $\mathbf{p}_j^{(t)}$ and velocities $\mathbf{v}_j^{(t)}$ of itself and its neighbors and receives exchanged messages $\{\mathbf{m}_j^{(t)} \mid \forall j \in \xi_i^{(t)}\}$, where $\mathbf{m}_j^{(t)}$ represents a learnable vector intended for communication and $\xi_i^{(t)}$ comprises agent i and its neighbors. Moreover, the global observation $\mathbf{o}_g^{(t)}$ can be expressed as $\mathbf{o}_g^{(t)} = \{(\mathbf{p}_i^{(t)}, \mathbf{v}_i^{(t)}) \mid \forall i \in \mathcal{N}\}$. Therefore, the global state $\mathbf{s}^{(t)} = (\mathbf{o}_g^{(t)}, \{\mathbf{m}_i^{(t)} \mid \forall i \in \mathcal{N}\}) \in \mathcal{S}$ encompasses local states $\mathbf{z}_i^{(t)} = (\mathbf{o}_i^{(t)}, \{\mathbf{m}_j^{(t)} \mid j \in \xi_i^{(t)}\})$ of all agents.

2) Action: Based on the local state $\mathbf{z}_i^{(t)}$, each agent determines its acceleration $\mathbf{u}_i^{(t)} = (u_{x_i}^{(t)}, u_{y_i}^{(t)}) \in \mathcal{U}$ according to its policy $\pi_{\theta_i}(\cdot \mid \mathbf{z}_i^{(t)})$ individually to accomplish assigned relative tasks.

3) Reward: We define the reward function as a weighted sum of multiple task-oriented and/or communication-related components, which are detailed in Section V.

D. Multi-Agent Proximal Policy Optimization

To steer all agents toward maximizing the discounted accumulated reward $\mathbb{E}[\sum_t \gamma^t \mathcal{R}^{(t)}]$, Multi-Agent Proximal Policy Optimization (MAPPO) [38] is employed as the base RL method, which combines single-agent PPO with the Centralized Training

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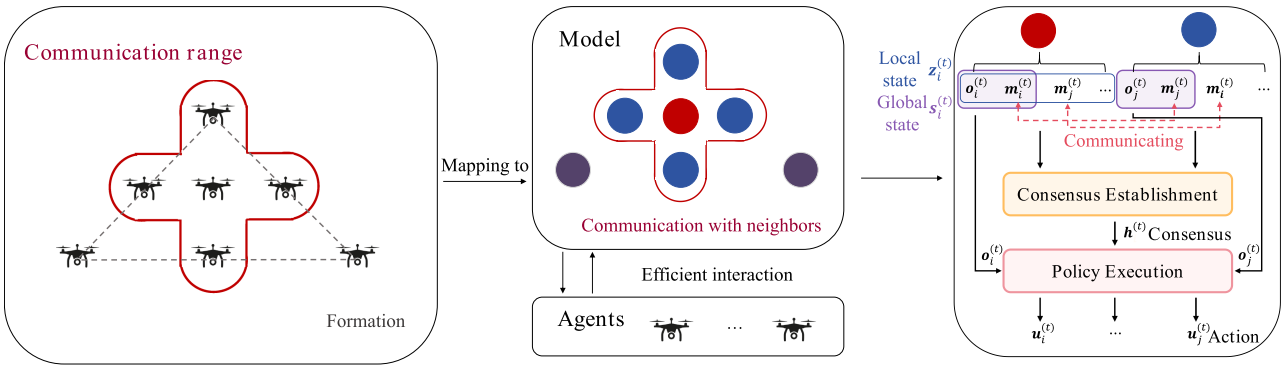


Fig. 2. Illustration of MARL information control.

and Decentralized Execution (CTDE) paradigm [24], aiming to learn both the individual policy $\pi_{\theta_i}(\cdot \mid \mathbf{z}_i^{(t)})$ for each agent i and the value function $V_\phi(\mathbf{s}^{(t)}) = \mathbb{E}_\pi[\sum_t \gamma^t \mathcal{R}^{(t)} | \mathbf{s}^{(t)}] : \mathcal{S} \to \mathbb{R}$, parameterized by θ_i and ϕ, respectively. Following the design of PPO, MAPPO retains the old versions of $\theta_{i,\text{old}}$ and ϕ_{old}, while $\theta_{i,\text{old}}$ is used to interact with the environment and accumulate the samples. Additionally, the parameters θ_i and ϕ are periodically updated to maximize the objective function,

$$J_{\pi_i}^{(t)}(\theta_i) = \min\left(\beta_i^{(t)}\hat{A}^{(t)},\ \text{clip}\left(\beta_i^{(t)}, 1-\varepsilon, 1+\varepsilon\right)\hat{A}^{(t)}\right),$$
$$J_V^{(t)}(\phi) = -\left(V_\phi\left(\mathbf{s}^{(t)}\right) - \left(\hat{A}^{(t)} + V_{\phi_{\text{old}}}\left(\mathbf{s}^{(t)}\right)\right)\right)^2, \quad (5)$$

where $\beta_i^{(t)} = \dfrac{\pi_{\theta_i}\left(\mathbf{u}_i^{(t)} | \mathbf{z}_i^{(t)}\right)}{\pi_{\theta_{i,\text{old}}}\left(\mathbf{u}_i^{(t)} | \mathbf{z}_i^{(t)}\right)}$ represents the importance ratio, ε denotes a hyperparameter, while the Generalized Advantage Estimation (GAE) $\hat{A}^{(t)} = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l \delta^{(t+l)}$ with the advantage estimate $\delta^{(t)} = \mathcal{R}^{(t)} + \gamma V_{\phi_{\text{old}}}(\mathbf{s}^{(t+1)}) - V_{\phi_{\text{old}}}(\mathbf{s}^{(t)})$. Consequently, the final optimization objective of MAPPO is given by

$$J_{\text{MAPPO}} = \mathbb{E}_{i,t}\left[J_{\pi_i}^{(t)}(\theta_i) + J_V^{(t)}(\phi) + \alpha H\left(\pi_{\theta_i}(\cdot \mid \mathbf{z}_i^{(t)})\right)\right], \quad (6)$$

where α is a coefficient and H represents the entropy function.

E. Problem Formulation

As illustrated in Fig. 2, our objective is to enable MARL-driven agents to maximize the global reward. Due to the partial observability of the global state $\mathbf{s}^{(t)}$ and the distinction over local states $\mathbf{z}_i^{(t)}$, it is essential to communicate essential information among neighboring nodes and infer some global consensus, thus making consistent actions in a decentralized manner. Many solutions [5], [6], [24] have been proposed in this area. Considering its performance superiority, ConsMAC [24] is taken into account in this manuscript, while the proposed solution is applicable to other works such as TarMAC [5] and MASIA [6].

Without loss of generality, at each time step t, each agent i needs to adaptively decide essential neighbor messages $\mathbf{m}_j^{(t)}$, the process of which is determined by a selection function $\mathcal{A}_i^{(t)}$. Afterward, the retained messages, together with the agent's local observation $\mathbf{o}_i^{(t)}$, are encoded into a latent feature $\mathbf{E}_{m_i}^{(t)}$.

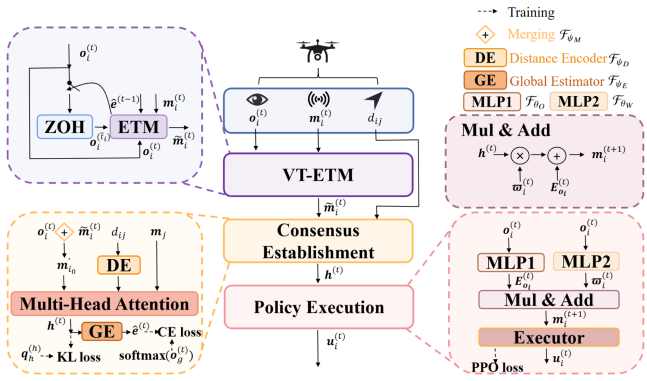


Fig. 3. The overall framework of CDE-GIB.

This embedding serves as the input to construct the consensus representation $\mathbf{h}^{(t)}$, which can be further used to interpret the global observation. In other words, if all agents share the same set of parameters [39] during training to enhance learning efficiency, expressed as $\pi_{\theta_i} = \pi_\theta \forall i \in \mathcal{N}$), it can be written as

$$\max_{\pi_\theta} \mathbb{E}_t\left[\sum_t \gamma^t \mathcal{R}^{(t)} \mid \pi_\theta\right],$$
$$\text{s.t.} \quad \mathbf{u}_i(t) = \pi_\theta\left(g\left(\mathcal{A}^{(t)}, \mathbf{o}_i^{(t)}\right)\right), \quad (7)$$

where the function g integrates the local observation $\mathbf{o}_i^{(t)}$ and event-filtered messages into a consensus representation $\mathbf{h}^{(t)}$ for decision making, which will be detailed in Section IV.

IV. The Framework of CDE-GIB

The overall framework of the proposed CDE-GIB, on top of ConsMAC [24], is shown in Fig. 3. In addition to the policy execution module, it also encompasses the VT-ETM module and GIB-based consensus establishment module, which evaluate the information importance based on current and historic observations and encode concise message representations for consensus inference, respectively. Beyond these functional pillars, the framework explicitly accounts for the propagation of consensus errors arising from aggregated multi-agent interactions.

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A. Fundamentals of ConsMAC

Generally, ConsMAC first leverages the combination of a Gated Recurrent Unit (GRU)-like memory module $\mathcal{F}_{\psi_M}$, parameterized by ψ_M, and positional encoding-based concatenation [40] to efficiently embed local observations and exchanged messages $\{\mathbf{m}_j^{(t)} \mid j \in \xi_i^{(t)}\}$ in $\mathbf{o}_i^{(t)}$. The mathematical formulation is provided in (8), shown at the bottom of this page where $||$ denotes the concatenation operation, $\mathbf{\Phi}_{d_j}^{(t)} = \sqrt{\frac{1}{D}}[\cos(w_1 L_2(\mathbf{p}_{i_j}^{(t)})), \ldots, \cos(w_D L_2(\mathbf{p}_{i_j}^{(t)}))]^\top$ with D learnable weights $\psi_D = [w_1, \ldots, w_D]$, alongside $L_2(\cdot)$ denoting the Euclidean norm and $\mathcal{A}(\cdot)$ indicating an event trigger function, which dynamically determines whether agent i receives the message from agent j at time t based on the VT-ETM. The detailed design of $\mathcal{A}(\cdot)$ is described in Section IV-B. Crucially, this formulation defines $\mathbf{E}_{m_i}^{(t)}$ as a composite representation, where the communication graph characterized by the distance encoder $\mathbf{\Phi}_{d_j}^{(t)}$, and the information flow represented by the memory module $\mathcal{F}_{\psi_M}$, are inextricably coupled. Notably, for each agent $i \in \mathcal{N}$, i_j represents the j-th nearest neighbor, while i_0 refers to the agent itself. Therefore, assuming the existence of a lossless channel from i to j, $\mathcal{A}(\cdot)$ becomes valid only if the occurrence of sending messages $\mathbf{m}_{i_j}^{(t)}$ from j to i.

Afterward, each agent then aggregates a latent vector $\mathbf{h}^{(t)}$[1] as

$$\mathbf{h}^{(t)} = \mathrm{MHA}_{\psi_A}\left(\mathbf{E}_{m_i}^{(t)}, \mathbf{E}_{m_i}^{(t)}, \mathbf{E}_{m_i}^{(t)}\right), \tag{9}$$

where MHA refers to a multi-head attention layer parameterized by ψ_A. On the basis, ConsMAC utilizes a global estimator $\mathcal{F}_{\psi_E}$, parameterized by ψ_E, to estimate the state embedding $\hat{\mathbf{e}}^{(t)} = \mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})$ in a supervised learning manner. Specifically, the Consensus Establishment (CE) loss function is computed as

$$\mathcal{L}_{\mathrm{CE}}(\Psi) = \mathbb{E}_t[\mathrm{MSE}(\mathbf{o}_g^{(t)} - \hat{\mathbf{e}}^{(t)})], \tag{10}$$

where $\Psi = [\psi_D, \psi_M, \psi_A, \psi_E]$. Through this process, the intermediate outputs of ConsMAC implicitly encode the global information, effectively establishing a consensus that captures the states of all agents. Meanwhile, the information flow also serves as an optimal message representation that encompasses all necessary details for the GIB. Besides, the message $\mathbf{m}_i^{(t+1)}$ for time-step $t+1$ will be computed as

$$\mathbf{m}_i^{(t+1)} = \mathbf{E}_{o_i}^{(t)} + \varpi_i^{(t)} \mathbf{h}^{(t)}, \tag{11}$$

where the embedding vector $\mathbf{E}_{o_i}^{(t)}$ and the communication information weight $\varpi_i^{(t)}$ can be obtained by Multi-Layer Percepton (MLP)-based encoders $\mathcal{F}_{\theta_O}$ and $\mathcal{F}_{\theta_W}$ as $\mathbf{E}_{o_i}^{(t)} = \mathcal{F}_{\theta_O}(\mathbf{o}_i^{(t)})$ and $\varpi_i^{(t)} = \mathcal{F}_{\theta_W}(\mathbf{o}_i^{(t)})$.

[1]Note that the subscript i is omitted in $\mathbf{h}^{(t)}$ since it represents a consensus embedding vector shared through local interactions, rather than an agent-specific feature.

Additionally, an executor $\mathcal{F}_{\theta_E}$ is employed to sample the final action output by calculating the mean of the Gaussian distribution as,

$$\mu_i^{(t)} = \mathcal{F}_{\theta_E}\left(\mathbf{m}_i^{(t+1)}\right), \quad \mathbf{u}_i^{(t)} \sim \mathrm{Normal}\left(\mu_i^{(t)}, \sigma^2\right), \tag{12}$$

where σ^2 represents the variance constant that introduces randomness to the agent's actions during exploration and gradually diminishes throughout the training process. As mentioned in Section III-D, in alignment with MAPPO, we consider $[\Theta, \Psi]$ as the parameters of the final policy π_{θ_i} in (5), where $\Theta = [\theta_O, \theta_W, \theta_E]$.

B. Variable Threshold-Event Triggered Mechanism

The practical design of VT-ETM involves two parts. Firstly, to effectively reduce the frequency of event triggering, we use a zero vector $\mathbf{o}_k^{(t)}$ as the label and apply Mean-Squared-Error (MSE) as the loss function to minimize the output of an event trigger function $\kappa(\cdot)$, which computes the semantic similarity between the predictive information induced by $\mathcal{F}_{\psi_T}$ and the inferred consensus $\hat{\mathbf{e}}^{(t-1)}$. Many similarity metrics can be adopted, such as cosine similarity, Manhattan distance, or Euclidean distance, depending on the characteristics of the data and the specific requirements of the task. Accordingly, the specific form of $\kappa(\cdot)$ is chosen from the following family of functions

$$\kappa(\mathbf{a}, \mathbf{b}) = \begin{cases} \dfrac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}, & \text{cosine similarity;} \\ -\sum_{i=1}^{K_2} |a_i - b_i|, & \text{neg. Manhattan;} \\ -\sqrt{\sum_{i=1}^{K_2} (a_i - b_i)^2}, & \text{neg. Euclidean,} \end{cases} \tag{13}$$

where $\mathbf{a} = \mathcal{F}_{\psi_T} \in \mathbb{R}^{K_2}$, and $\mathbf{b} = \hat{\mathbf{e}}^{(t-1)} \in \mathbb{R}^{K_2}$.

When the predictive information from an agent's local observation closely matches the inferred consensus, it signifies that the agent's local perspective aligns well with the current state, thereby demonstrating the reliability of its observation information. This consistency suggests that the agent possesses valuable and trustworthy exchanged messages that are likely to contribute meaningfully to the consensus establishment. Consequently, it ensures $\kappa(\cdot)$ to become as small as possible (i.e., the minimization of unnecessary triggering). Mathematically, the formula can be shown as

$$\begin{aligned} &\mathcal{L}_{\text{VT-ETM}}(\psi_T) \\ &= \mathbb{E}_{i,t}\left[\mathrm{MSE}(\kappa\left\{\mathcal{F}_{\psi_T}\left(\mathbf{o}_i^{(t)}, \mathbf{o}_i^{(\tilde{t}_i)}\right), \hat{\mathbf{e}}^{(t-1)}\right\} - \mathbf{o}_k^{(t)})\right], \end{aligned} \tag{14}$$

where $\mathcal{F}_{\psi_T}$ denotes an MLP parameterized by ψ_T to process the agent i's observation $\mathbf{o}_i^{(t)}$ and its historical local observation $\mathbf{o}_i^{(\tilde{t}_i)}$ at the last trigger moment $\tilde{t}_i$ that is memorized by Zero-Order Hold (ZOH) [41]. Note that based on a function

$$\mathbf{E}_{m_i}^{(t)} = \left[\mathcal{F}_{\psi_M}\left([\mathbf{m}_{i_0}^{(t)} \,\|\, \mathbf{o}_i^{(t)}]\right) \,\|\, \mathbf{\Phi}_{d_0}^{(t)}, \ldots, \left(\mathcal{A}_j^{(t)} \,\|\, \mathbf{\Phi}_{d_j}^{(t)}\right), \ldots, \left(\mathcal{A}_N^{(t)} \,\|\, \mathbf{\Phi}_{d_N}^{(t)}\right)\right]^\top, \tag{8}$$

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$\kappa(\cdot)$, each agent i maintains a sequence of event-triggering times $\mathcal{T}_i$. When the agent determines that its information is semantically useful to other agents as indicated by $\mathcal{A}^{(t)}$, it will add the corresponding time t to the set $\mathcal{T}_i$. Correspondingly, $\tilde{t}_i = \arg\min_{\tau \in \mathcal{T}_i} \{t - \tau\} \in \mathcal{T}_i$. During the centralized training, we add ψ_T to $\Psi = [\psi_D, \psi_M, \psi_A, \psi_E, \psi_T]$[2] for parameter updating.

On the other hand, during decentralized execution, we introduce an exponential function that decreases over time, namely $G_{\text{threshold}} = c\zeta^t$, where $c > 0$ and $0 < \zeta < 1$, as the variable threshold. In other words, for agent i

$$\mathcal{A}_i^{(t)} \text{ is } \begin{cases} \texttt{VALID}, & \kappa\left\{\mathcal{F}_{\psi_T}\left(\mathbf{o}_i^{(t)}, \mathbf{o}_i^{(\tilde{t}_i)}\right), \hat{\mathbf{e}}^{(t-1)}\right\} > G_{\text{threshold}}; \\ \texttt{VOID}, & \kappa\left\{\mathcal{F}_{\psi_T}\left(\mathbf{o}_i^{(t)}, \mathbf{o}_i^{(\tilde{t}_i)}\right), \hat{\mathbf{e}}^{(t-1)}\right\} \leq G_{\text{threshold}}. \end{cases} \tag{15}$$

Exploiting a time-decreasing threshold $G_{\text{threshold}}$ aligns with the branch-out approaches used to mitigate accumulative errors during model rollouts [42], progressively compromising the accuracy of predictions. Therefore, an adaptive threshold that encourages increasingly frequent updates can naturally counteract the gradually enlarged errors. Finally, if $\mathcal{A}_i^{(t)}$ is `VOID`, neighboring agents will be unable to receive any exchanged messages from agent i. In this case, instead of using the memorized message in [20], the corresponding elements in (8) will be replaced by an all-zero vector. Such a setting contributes to allow agents receiving messages infrequently to avoid over-reliance on outdated memorized messages that may hinder their ability to make timely decisions.

Definition 1. (Zeno Behavior [43]): For an agent pair (i, j), Zeno behavior occurs when there emerges an infinite number of triggering times until a finite time.

Notably, in the RL framework, the inherent simulation time step enforces a minimum inter-event interval, which effectively mitigates the risk of such behavior.

In terms of computational complexity, VT-ETM introduces a lightweight MLP forward computation $\mathcal{F}_{\psi_T}$ and a similarity evaluation $\kappa(\cdot)$ for each agent at every time step. This yields a per-agent complexity of $\mathcal{O}(d_T + d)$, where d_T denotes the number of parameters in the MLP $\mathcal{F}_{\psi_T}$, and d is the dimension of the feature vector involved in the similarity computation $\kappa(\cdot)$. Since $d \ll d_T$ in practice, the cost of similarity computation is negligible compared to the MLP forward computation. Therefore, the total complexity across N agents is $\mathcal{O}(N \cdot d_T)$ per step. Furthermore, by selectively suppressing communication, VT-ETM reduces the computational burden associated with message passing and aggregation, achieving a favorable trade-off between coordination quality and efficiency.

C. Graph Information Bottleneck for the Communication Graph and Information Flow Optimization

As mentioned above, MAGI [14] generates excessive communication volume due to the cumbersome interactions with all agents within a specified range. Moreover, the algorithm suffers from high computational complexity caused by the separate compression of the communication graph and information flow. To address this challenge, we propose the GIB-based joint optimization. Specifically, different from MAGI [14], which considers the triplet ⟨`Feature`, `Communication/Graph Information`, `Explicit Action`⟩, our method incorporates a novel triplet ⟨`Feature`, `Implicit Consensus`, `Global Observation`⟩ to infer the consensus from information flow and the communication graph simultaneously.

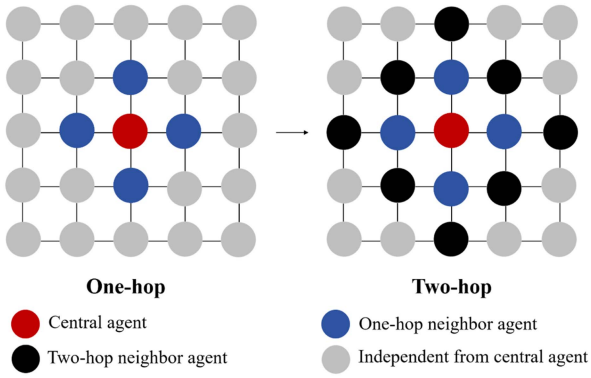


Fig. 4. Visualization of local-dependence assumption.

Beforehand, for graph-structured agent features, we introduce a local-dependence assumption [13] to avoid explicitly requiring input data to be i.i.d.

Assumption 1. (Local-Dependence Assumption): For each agent i, given the neighbor-related agents within a certain number of hops, the features of the remaining agents are considered independent of the feature of agent i, as depicted in Fig. 4 .

The local-dependence assumption has been extensively adopted in existing literature [13], [14] and has shown robust empirical validity across varying neighborhood ranges, as further examined in Fig. 12(c).

Contingent on Assumption 1, we discuss the implementations of the GIB-based joint compression of the communication graph and information flow towards a more compact consensus representation. Without loss of generality, the input feature data for the communication learning mechanism based on GNNs can be universally expressed as $\mathcal{D} = (\mathcal{E}, \mathcal{H})$, where $\mathcal{D}$ denotes integrated information that combines the communication graph and node-level consensus. Therefore, our primary objective is to compress the consensus $\mathbf{h}^{(t)}$ from $\mathbf{E}_{m_i}^{(t)}$, while promoting consensus $\mathbf{h}^{(t)}$ to closely approximate the target global observation label $\mathbf{o}_g^{(t)}$. Mathematically, we minimize the following objective function

$$\mathcal{L}_{\text{GIB}} = -I(\mathbf{h}^{(t)}; \mathbf{o}_g^{(t)}) + \eta I(\mathbf{h}^{(t)}; \mathbf{E}_{m_i}^{(t)}). \tag{16}$$

However, it is quite challenging to compute the mutual information between two high-dimensional continuous random variables $\mathbf{h}^{(t)}$ and $\mathbf{o}_g^{(t)}$ as well as $\mathbf{h}^{(t)}$ and $\mathbf{E}_{m_i}^{(t)}$. To approximate this intractable computation, we employ the variational method inspired by Variational Information Bottleneck (VIB) [44].

[2]For simplicity of representation, we slightly abuse the notations here.

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Lemma 1 (Nguyen, Wainright & Jordan's bound [13]): For two random variables X and Y,

$$I(Y;X) \geq 1 + \mathbb{E}_{p(Y)}\left[\log \frac{\prod_{i\in\mathcal{N}} p_1(Y_i \mid X_i)}{p_2(Y)}\right] - \mathbb{E}_{p(Y)p(X)}\left[\frac{\prod_{i\in\mathcal{N}} p_1(Y_i \mid X_i)}{p_2(Y)}\right], \tag{17}$$

where $\mathcal{N}$ denotes the pool of nodes.

Based on Lemma 1, we have the following theorem.

Theorem 1: The GIB can be bounded by

$$\mathcal{L}_{\text{GIB}} \leq \mathbb{E}_t\left[D_{\text{KL}}\left(p(\mathbf{o}_g^{(t)}) \parallel p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))\right)\right] + \eta\mathbb{E}_{p(\mathbf{E}_{m_i}^{(t)})}\left[D_{\text{KL}}\left(p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)}) \parallel q_{\mathbf{h}}(\mathbf{h}^{(t)})\right)\right], \tag{18}$$

where $q_{\mathbf{h}}(\mathbf{h}^{(t)})$ denotes a probability function sharing the same variable space as $p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)})$.

Proof:

$$\begin{aligned} & I\left(\mathbf{h}^{(t)};\mathbf{o}_g^{(t)}\right) \\ \overset{(a)}{\geq} & 1 + \mathbb{E}_{p(\mathbf{o}_g^{(t)})}\left[\log \frac{p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))}{p\left(\mathbf{o}_g^{(t)}\right)}\right] \\ & - \mathbb{E}_{p(\mathbf{o}_g^{(t)})p(\mathbf{h}^{(t)})}\left[\frac{p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))}{p\left(\mathbf{o}_g^{(t)}\right)}\right] \\ \overset{(b)}{\geq} & -\mathbb{E}_{p(\mathbf{o}_g^{(t)})}\left[\log \frac{p\left(\mathbf{o}_g^{(t)}\right)}{p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))}\right] \\ = & -\mathbb{E}_t\left[D_{\text{KL}}\left(p(\mathbf{o}_g^{(t)}) \parallel p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))\right)\right], \end{aligned} \tag{19}$$

where the inequality (a) uses Lemma 1 and the inequality (b) is derived from $1 - \mathbb{E}_{p(\mathbf{o}_g^{(t)})p(\mathbf{h}^{(t)})}[\frac{p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)}))}{p(\mathbf{o}_g^{(t)})}] > 0$.

On the other hand,

$$\begin{aligned} & I(\mathbf{h}^{(t)};\mathbf{E}_{m_i}^{(t)}) \\ = & \iint p(\mathbf{h}^{(t)},\mathbf{E}_{m_i}^{(t)}) \log \frac{p(\mathbf{h}^{(t)},\mathbf{E}_{m_i}^{(t)})}{p(\mathbf{h}^{(t)})p(\mathbf{E}_{m_i}^{(t)})}\, d\mathbf{h}^{(t)}\, d\mathbf{E}_{m_i}^{(t)} \\ = & \iint p(\mathbf{E}_{m_i}^{(t)})p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)}) \log \frac{p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)})}{p(\mathbf{h}^{(t)})}\, d\mathbf{h}^{(t)}\, d\mathbf{E}_{m_i}^{(t)} \\ \overset{(c)}{\leq} & \iint p(\mathbf{E}_{m_i}^{(t)})p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)}) \log \frac{p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)})}{q_{\mathbf{h}}(\mathbf{h}^{(t)})}\, d\mathbf{h}^{(t)}\, d\mathbf{E}_{m_i}^{(t)} \\ = & \mathbb{E}_{p(\mathbf{E}_{m_i}^{(t)})}\left[D_{\text{KL}}\left(p(\mathbf{h}^{(t)} \mid \mathbf{E}_{m_i}^{(t)}) \parallel q_{\mathbf{h}}(\mathbf{h}^{(t)})\right)\right], \end{aligned} \tag{20}$$

where the inequality (c) originates from Gibbs' inequality, namely $p\log p \geq p\log q$, with equality if and only if p and q are the same distribution. To further estimate $p(\mathbf{h}^{(t)})$, we treat $q_{\mathbf{h}}(\mathbf{h}^{(t)})$ as the variational approximation. ■

Compared to (10), Theorem 1 unveils the impact of GIB on the consensus-building module. In a nutshell, the loss function of CDE-GIB can be expressed as,

$$\mathcal{L}_{\text{CDE-IB}}(\Theta,\phi,\Psi) = -J_{\text{MAPPO}} + \mathcal{L}_{\text{CE}} + \varrho\mathcal{L}_{\text{VT-ETM}} + \rho\mathcal{L}_{\text{GIB}}. \tag{21}$$

However, it remains difficult to compute the Kullback–Leibler (KL) divergence directly during training. Fortunately, we have the following lemma.

Lemma 2 (Ref. [45]): Considering $f(\mathbf{x})$ and $g(\mathbf{x})$ ($\mathbf{x} \in \mathbb{R}^K$) as Gaussian distributions, that is,

$$f(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m),\ g(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l), \tag{22}$$

where $\boldsymbol{\mu} = [\mu^1,\ldots,\mu^K]$[3] and $\boldsymbol{\Sigma} = \text{diag}[(\sigma^1)^2,\ldots,(\sigma^K)^2]$, the KL divergence between these two distributions can be computed as

$$\begin{aligned} & \mathbb{E}\left[D_{\text{KL}}\left(f(\mathbf{x}) \parallel g(\mathbf{x})\right)\right] \\ = & \mathbb{E}\left[\sum_{k=1}^{K}\left(\log\frac{\sigma_m^k}{\sigma_l^k} + \frac{\left(\sigma_m^k\right)^2 + \left(\mu_m^k - \mu_l^k\right)^2}{2\left(\sigma_l^k\right)^2} - \frac{1}{2}\right)\right]. \end{aligned} \tag{23}$$

Contingent on the following Assumption 2, we can have a corollary to faciliate the computations of GIB, which lays the very foundation for computing GIB across batches.

Assumption 2 (Consistent with Ref. [12]): Taking a batch of collected data, we assume $\mathbf{o}_g^{(t)}$, $\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})$, and $\mathbf{h}^{(t)}$ satisfy the following Gaussian distributions,

$$\begin{aligned} p(\mathbf{o}_g^{(t)}) &= \mathcal{N}(\boldsymbol{\mu}_{\mathbf{o}_g^{(t)}}, \boldsymbol{\Sigma}_{\mathbf{o}_g^{(t)}}), \\ p(\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})) &= \mathcal{N}(\boldsymbol{\mu}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}, \boldsymbol{\Sigma}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}), \\ p(\mathbf{h}^{(t)}) &= \mathcal{N}(\boldsymbol{\mu}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}, \boldsymbol{\Sigma}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}), \end{aligned} \tag{24}$$

where $\boldsymbol{\mu}_{\mathbf{o}_g^{(t)}}$, $\boldsymbol{\mu}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}$, $\boldsymbol{\sigma}_{\mathbf{o}_g^{(t)}}$, $\boldsymbol{\sigma}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})} \in \mathbb{R}^{K_1}$, and $\boldsymbol{\mu}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}$, $\boldsymbol{\sigma}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}} \in \mathbb{R}^{K_2}$.

The Gaussian approximation serves as a common and necessary simplification to ensure stable variational optimization [12], [44], while maintaining sufficient expressiveness in practice.

Corollary 1: Contingent on Assumption 2, if the variational approximation $q_{\mathbf{h}}(\mathbf{h}^{(t)}) = \mathcal{N}(\mathbf{0},\mathbf{I})$, the GIB bound can be computed as (25), as shown at the bottom of the next page.

The corollary can be easily obtained by applying Assumption 2 and Lemma 2 in Theorem 1. In summary, the training procedure for CDE-GIB is presented in Algorithm 1.

D. Investigating the Impact of Noise in GNN-Based Communication Paradigm on GIB Optimization

In this part, we derive a noise-perturbed version of the GIB, wherein the mutual information between the agent's embedding representation $\mathbf{E}_{m_i}^{(t)}$ and consensus $\mathbf{h}'^{(t)}$, as well as between the global observation label $\mathbf{o}_g^{(t)}$ and consensus $\mathbf{h}'^{(t)}$, are recalculated, by regarding the arising cumulative error present in the consensus-learning process as some kind of noise. The revised GIB can better adapt to perturbations affecting GNN nodes [17] and errors accumulated during the consensus inference process.

[3]For simplicity of representation, we omit the subscript m and l for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ here.

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Assumption 3: The node feature attributes $\mathbf{h}^{(t)}$ in the GNN-based communication paradigm are perturbed by independent, additive Gaussian noise $\mathbf{c}^{(t)}$ [46]. Formally, the noise-corrupted feature is defined as the linear superposition of original features and noise: $\mathbf{h}^{'(t)} = \mathbf{h}^{(t)} + \mathbf{c}^{(t)}$.

It is important to note that although the node-level noise introduced in Assumption 3 is modeled as additive and independent, the actual process of consensus establishment exhibits a temporally recursive structure that induces error accumulation. Specifically, the consensus $\mathbf{h}^{'(t)}$, already affected by noise, is used to compute the agent's message at time $t+1$ in (11), which subsequently contributes to the next-step consensus. This established consensus is again perturbed by additive noise in the GNN-based paradigm. To mitigate this long-horizon degradation, VT-ETM reduces the propagation of noisy messages via selective communication, while the revised GIB explicitly accounts for additive noise within its information-theoretic formulation.

Lemma 3: Building upon Assumption 3, and taking a batch of collected data, we assume $\mathcal{F}_{\psi_E}(\mathbf{h}^{'(t)})$, and $\mathbf{h}^{'(t)}$ under noise perturbations satisfy the following Gaussian distributions,

$$\begin{aligned} p(\mathcal{F}_{\psi_E}(\mathbf{h}^{'(t)})) &= \mathcal{N}\left(\boldsymbol{\mu}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{'(t)})}, \boldsymbol{\Sigma}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{'(t)})}\right) \\ &= \mathcal{N}(\boldsymbol{\mu}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}, \boldsymbol{\Sigma}_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}) + \iota\mathcal{N}(\mathbf{0}, \sigma_c^2\mathbf{I}), \\ p(\mathbf{h}^{'(t)}) &= \mathcal{N}(\boldsymbol{\mu}_{\mathbf{h}'^{(t)}|\mathbf{E}_{m_i}^{(t)}}, \boldsymbol{\Sigma}_{\mathbf{h}'^{(t)}|\mathbf{E}_{m_i}^{(t)}}) \\ &= \mathcal{N}(\boldsymbol{\mu}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}, \boldsymbol{\Sigma}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}) + \mathcal{N}(\mathbf{0}, \sigma_c^2\mathbf{I}), \end{aligned} \tag{26}$$

where $\boldsymbol{\mu}_{\mathcal{F}_{\psi_E}(\mathbf{h}'^{(t)})}$, $\boldsymbol{\sigma}_{\mathcal{F}_{\psi_E}(\mathbf{h}'^{(t)})} \in \mathbb{R}^{K_1}$, $\boldsymbol{\mu}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{'(t)}}$, $\boldsymbol{\sigma}_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{'(t)}} \in \mathbb{R}^{K_2}$, and ι represents a modulation factor that scales the noise component after its propagation through the MLP.

Corollary 2: Based on Assumption 2 and Assumption 3, when the variational approximation $q_{\mathbf{h}}(\mathbf{h}^{'(t)}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, the GIB bound under noise perturbations can be computed as (27), shown at the bottom of the page.

The corollary is readily derived from Theorem 1 by invoking Lemma 2 and Lemma 3, alongside the nice property of summarizing two Gaussian distributions [47]. In conclusion, the training procedure for CDE-GIB is also detailed in Algorithm 1. Notably, for convenience of representation, we denote the CDE-GIB with the modified loss function in (27) as CDE-GIB-n.

Lemma 4 (Policy Iteration Convergence [48]): If the value function is bounded and the policy update ensures monotonic improvement, then the policy iteration process converges to an optimal policy π^*.

Theoretically, the framework also ensures convergence by meeting the two conditions in Lemma 4. Value boundedness stems from $\gamma \in (0,1)$ and bounded rewards, and monotonic improvement is maintained via encouraging the new policy to favor actions with higher Q-values under the guidance of the critic.

To better assess the computational efficiency of the proposed GIB, we provide a brief complexity analysis here. The loss functions in (25) and (27) are derived from the closed-form KL divergence between multivariate Gaussian distributions. These calculations involve only element-wise arithmetic operations over K_1 and K_2 dimensional vectors. Therefore, the overall complexity across n agents is $\mathcal{O}(K_1 + K_2)$, which indicates the computational tractability of the proposed formulation.

V. Simulation Settings and Results

A. Simulation Settings

In this section, we evaluate the performance of CDE-GIB in terms of executing the decentralized formation control task [24] in the multi-agent particle environment [23]. Notably, we use

$$\begin{aligned} \mathcal{L}_{\text{GIB}} \leq \mathbb{E}\left[\sum_{k=1}^{K_1}\left(\log\frac{\sigma^k_{\mathbf{o}_g^{(t)}}}{\sigma^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}} + \frac{\left(\sigma^k_{\mathbf{o}_g^{(t)}}\right)^2 + \left(\mu^k_{\mathbf{o}_g^{(t)}} - \mu^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}\right)^2}{2\left(\sigma^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}\right)^2}\right)\right] \\ + \mathbb{E}\left[\sum_{k=1}^{K_2}\left(\log\sigma^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}} + \frac{\left(\sigma^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}\right)^2 + \left(\mu^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}\right)^2}{2}\right)\right] - 1. \end{aligned} \tag{25}$$

$$\begin{aligned} \mathcal{L}^{\text{noise}}_{\text{GIB}} \leq \mathbb{E}\left[\sum_{k=1}^{K_1}\left(\log\frac{\sigma^k_{\mathbf{o}_g^{(t)}}}{\sqrt{\left(\sigma^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}\right)^2 + \iota^2\left(\sigma^k_c\right)^2}} + \frac{\left(\sigma^k_{\mathbf{o}_g^{(t)}}\right)^2 + \left(\mu^k_{\mathbf{o}_g^{(t)}} - \mu^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}\right)^2}{2\left(\sigma^k_{\mathcal{F}_{\psi_E}(\mathbf{h}^{(t)})}\right)^2 + 2\iota^2\left(\sigma^k_c\right)^2}\right)\right] \\ + \mathbb{E}\left[\sum_{k=1}^{K_2}\left(\log\sqrt{\left(\sigma^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}\right)^2 + \left(\sigma^k_c\right)^2} + \frac{\left(\sigma^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}\right)^2 + \left(\sigma^k_c\right)^2 + \left(\mu^k_{\mathbf{h}^{(t)}|\mathbf{E}_{m_i}^{(t)}}\right)^2}{2}\right)\right] - 1. \end{aligned} \tag{27}$$

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Algorithm 1: The Training of CDE-GIB.

Initialize: Inference steps T_{infer}, training steps T_{train}, variance constant σ, the actor and critic network with random parameters Θ, Ψ, ϕ and the replay memory $\mathcal{B} \leftarrow \varnothing$;

for each training episode **do**
  Clone $\Theta_{\text{old}} \leftarrow \Theta$, $\Psi_{\text{old}} \leftarrow \Psi$, $\phi_{\text{old}} \leftarrow \phi$;
  Initialize the environment with N agents;
  for $t = \{1, \ldots, T_{\text{infer}}\}$ **do**
    for each agent i **do**
      $\mathbf{z}_i^{(t)} \leftarrow$ obtains a local state $\mathbf{z}_i^{(t)} = (\mathbf{o}_i^{(t)}, \{\mathbf{m}_j^{(t)} \mid j \in \xi_i^{(t)}\})$;
      $\mathcal{A}_i \leftarrow$ calculates triggering behavior $\mathcal{A}_i$ by (15);
      $\mathbf{E}_{m_i}^{(t)} \leftarrow$ generates encoded information $\mathbf{E}_{m_i}^{(t)}$ by (8);
      $\mathbf{h}^{(t)}, \mathbf{m}_i^{(t+1)}, \mathbf{u}_i^{(t)} \leftarrow$ establishes the consensus $\mathbf{h}^{(t)}$ by (9) and computes the message $\mathbf{m}_i^{(t+1)}$ by (11) to sample an action $\mathbf{u}_i^{(t)} \sim \text{Normal}(\mu_{\text{i, old}}^{(t)}, \sigma^2)$ by (12);
    end for
    $r^{(t)}, V_{\phi_{\text{old}}}(\mathbf{s}^{(t)}), \mathbf{s}^{(t+1)} \leftarrow$ obtain the reward $r^{(t)}$, state value $V_{\phi_{\text{old}}}(\mathbf{s}^{(t)})$ and $\mathbf{s}^{(t+1)}$;
    Store $(\mathbf{s}^{(t)}, \mathbf{h}^{(t)}, \mathbf{u}^{(t)}, r^{(t)}, \mathbf{s}^{(t+1)})$ in $\mathcal{B}$;
  end for
  for $t = \{1, \ldots, T_{\text{train}}\}$ **do**
    Sample a batch of B_n from $\mathcal{B}$;
    Update Θ, Ψ, ϕ according to (21) via Adam optimizer, where the GIB term can be represented by either (25) or (27), conditional on the consideration of noise.
  end for
end for

cosine similarity during computing κ in (14). Additionally, the reward function $\mathcal{R}^{(t)}$ is defined as a weighted combination of task-oriented reward $\mathcal{R}_t^{(t)}$ and the event-triggered reward $\mathcal{R}_m^{(t)}$. Specifically, consistent with [24], the task-oriented reward $\mathcal{R}_t^{(t)}$, which contains the formation completeness, individual navigation distance, and penalty on collision arising from decentralized control, evaluates the agents' efficacy in executing specified tasks. Meanwhile, the event-triggered reward $\mathcal{R}_m^{(t)}$ imposes a penalty on agents for transmitting information. Mathematically,

$$\mathcal{R}^{(t)} = \omega_k \mathcal{R}_t^{(t)} + \omega_m \mathcal{R}_m^{(t)}, \tag{28}$$

where the coefficients ω_t, ω_m represent the corresponding weights. We also evaluate the distributed consensus establishment method ConsMAC [24], the attention-based message aggregation approach TarMAC [5], and the state-of-the-art supervised learning-based information extraction algorithm MASIA [6] as baselines.

During the experiments, the value function network is implemented as a three-layer MLP, while other modules utilize two-layer MLPs. All hidden layers and message dimensions are uniformly set to 256, with the attention mechanism employing a four-head self-attention architecture. The training protocol combines batch data collection (20 parallel threads generating 100-step per episode) with iterative optimization (15 gradient updates per episode with a learning rate 1×10^{-4}), incorporating MAPPO-standard techniques including orthogonal initialization, gradient clipping, and value normalization to ensure training stability. The communication channel is modeled as a Rician fading channel, with the Rice factor K_L set to 3.5333 [34], corresponding to a typical UAV operational altitude of 10 m. We evaluate agents' performance after each training episode to track the learning progress. The key parameters are summarized in Table III.

TABLE III
THE KEY PARAMETER SETTINGS OF THE ENVIRONMENT

Environment Parameters	Notation	Value
Inference steps	T_{infer}	100
Training steps	T_{train}	15
Number of episodes	T	250
Number of threads	T_{thread}	20
Number of UAVs	N	7
Maximum observation distance	δ_{obs}	3 m
Destination	Δ_p	$(0, 10)$ m
Discount factor	γ	0.8
GAE factor	λ	0.95
The range of acceleration	$\mathbf{u}_i^{(t)}$	$[-0.5, 0.5]$ m/s^2
The range of position	$\mathbf{p}_i^{(0)}, \mathbf{p}_j^{(0)}$	$[-2, 2]$ m
Reward function coefficients	ω_k, ω_m	1, 0.1
Loss function coefficients	ϱ, ρ	1,1
Dimension of the global observation	K_1	28
Dimension of hidden layers	K_2	256
Rice factor	K_L	3.5333
Learning rate	lr	1×10^{-4}
Scaling factor for the variable threshold	c	1
Exponential decay rate of the threshold	ζ	0.6

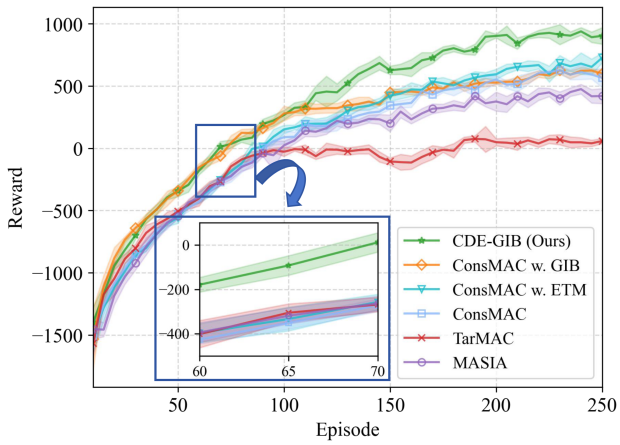


Fig. 5. Learning curves of consensus algorithms with and without GIB optimization.

B. *Effectiveness of GIB*

Fig. 5 validates the superiority of CDE-GIB over ConsMAC [24], TarMAC [5], and MASIA [6]. All experiments are conducted under noiseless conditions with 7 agents and random seed 1. Under these settings, Fig. 6 presents the trajectory snapshots of CDE-GIB-empowered agents to achieve a formation

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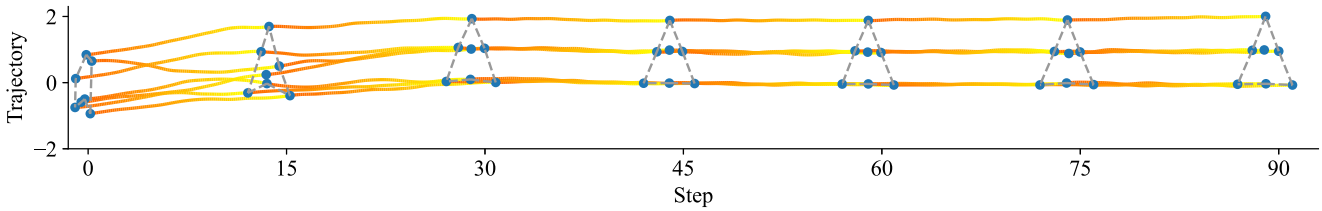


Fig. 6. Trajectory snapshots from Step 1 to 90 in Episode 240.

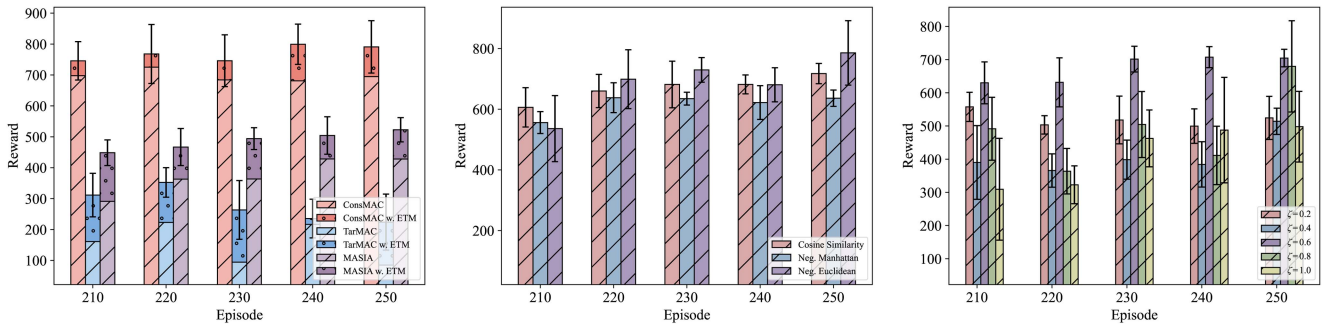


Fig. 7. Generality evaluation of VT-ETM. (a) Performance comparison of consensus algorithms with and without ETM. (b) Performance comparison of different event trigger functions. (c) Communication-performance trade-off analysis under varying event trigger thresholds.

as illustrated in Fig. 2. On this basis, we conduct the following analyses.

Fig. 5 further demonstrates that the incorporation of GIB leads to faster convergence compared to the baseline ConsMAC. Moreover, it achieves higher task rewards than TarMAC and MASIA as well.

Discussions: The effectiveness of GIB arises from its principled integration of IB theory into decentralized multi-agent coordination. By rigorously enforcing a minimal, sufficient compression paradigm, GIB fundamentally redefines how agents distill emergent consensus from high-dimensional messages and observations. In contrast, while TarMAC demonstrates the accelerated convergence rate during training, it achieves suboptimal final performance compared to ConsMAC. This limitation likely arises from the lack of explicit guidance over the content of communication, resulting in the transmission of uninterpretable information. Furthermore, MASIA solely transmits the raw observation, which provides insufficient guidance for agent decision-making. In addition, the learning curve of ConsMAC w. ETM is generally higher than that of ConsMAC, indicating improved communication effectiveness even at early learning stages.

C. Generality of VT-ETM

This part aims to validate the universal applicability of VT-ETM systematically. All experiments in this subsection employ a consistent configuration of 7 agents with random seed 1 under noiseless conditions.

1) Compatibility With Different Consensus Algorithms: To rigorously assess its compatibility and performance consistency, we integrate VT-ETM into multiple baseline algorithms [5], [6], [24]. Crucially, no algorithmic-specific modifications are introduced during the integration of VT-ETM, ensuring a fair evaluation of its inherent generality. Fig. 7(a) demonstrates the corresponding results under different episodes. A clear trend emerges that the adoption of VT-ETM leads to notable performance improvement for all methods.

2) Compatibility With Different Event Trigger Functions: To address the generality of the event trigger function κ, we evaluate three distinct formulations within the VT-ETM framework according to (13). As shown in Fig. 7(b), all three trigger functions achieve competitive performance. This demonstrates VT-ETM's flexibility in accommodating different similarity metrics while maintaining robust performance.

3) Impact of the Threshold Decay Rate: We evaluate the communication-performance tradeoff induced by the variable-threshold mechanism under different values of ζ. As shown in Fig. 7(c), performance exhibits a non-monotonic trend, with intermediate values striking a better balance between communication efficiency and coordination performance. In contrast, too small thresholds cause excessive communication and unstable coordination, while too large thresholds suppress communication and impair cooperation.

4) Comparison With Existing Event-Triggering Schemes: To further validate the effectiveness of VT-ETM, we compare it with representative event-triggering mechanisms:

- Observation-based triggering [18], [19], where the event trigger function is constructed solely from current local observations without incorporating historical information.

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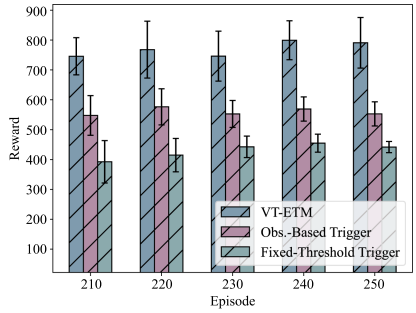


Fig. 8. Performance comparison with existing event-triggering schemes.

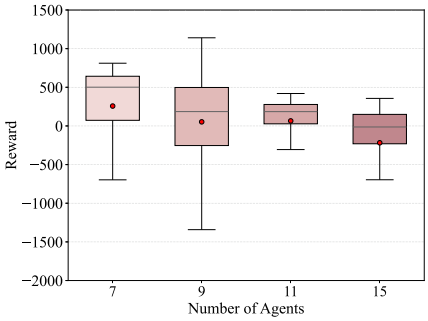


Fig. 9. Scalability performance with varying numbers of agents.

- Fixed-threshold triggering [21], where the event trigger function is constructed based on predictive information but the threshold remains constant during execution.

Fig. 8 indicates that VT-ETM consistently achieves the highest final reward among all compared methods, reflecting its stronger capability in leveraging predictive information for effective coordination.

Discussions: The performance improvement of VT-ETM stems from its dual emphasis on historical context integration and dynamic importance evaluation. This balanced design is further reinforced by its threshold variability, which adaptively adjusts triggering criteria to minimize cumulative prediction errors during extended model rollouts. The framework's compatibility with multiple event trigger functions also underscores its generalizability, as different similarity metrics can be seamlessly incorporated without compromising performance.

D. Scalability and Formation Diversity of CDE-GIB

Fig. 9 evaluates the proposed CDE-GIB under joint changes of population size and formation topology. As the number of agents increases from 7 to 9, 11 and 15, the formation topology is adapted in tandem with the population size, employing a distinct, scale-aware formation at each N to reflect practical deployment. All experiments are conducted in noiseless settings with a random seed of 1. Specifically, while the median reward decreases from 501.28 ($N = 7$) to 185.48 ($N = 9$), it stabilizes at 185.09 ($N = 11$) with significantly reduced variability, as evidenced by a narrower IQR of 250.45. When further scaling to 15 agents, the median slightly decreases to -13.45 with an IQR of 380.44.

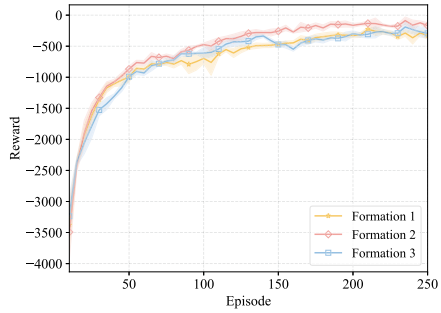


Fig. 10. Performance of CDE-GIB across diverse 3D formation configurations.

Discussions: The consistent or improved performance under larger-scale topologies verifies the superior scalability of CDE-GIB. The notably reduced variability at N=11 reflects the model's capacity to re-stabilize under denser communication graphs, while the acceptable performance at N=15 indicates resilience to increased coordination uncertainty. This favorable scalability can be attributed to the communication-efficient design of the VT-ETM and GIB modules.

E. Extension to Three-Dimensional Scenarios

To address the concern regarding the fixed-altitude assumption in the primary experiments, we further evaluate the proposed CDE-GIB framework in a fully three-dimensional formation control scenario. Specifically, we construct three representative 3D formations with varying vertical distributions, as illustrated in Fig. 11. The results shown in Fig. 10 demonstrate that CDE-GIB successfully stabilizes all three 3D formations and achieves consistent convergence behavior.

Discussions: Without structural modifications to the communication architecture or training procedure, the framework effectively scales to 3D coordination scenarios. This extension confirms the robustness and scalability of the proposed method under more realistic and dynamically complex environments.

F. Impact of Hyperparameter Configurations

This experiment systematically assesses the efficacy of the proposed CDE-GIB framework by leveraging boxplot visualizations to quantify the statistical distribution of episodic rewards during the post-convergence phase (Episodes 201–250, a period marked by decelerated convergence and a progressively flattening reward curve). All experiments are conducted using a consistent configuration of 7 agents, with a random seed 1 and under noiseless conditions.

As presented in Fig. 12(a), the median reward of CDE-GIB exhibits a pronounced non-monotonic relationship, which suggests that a properly calibrated event-triggering weight can better balance communication efficiency and policy learning. Fig. 12(b) shows that increasing the GIB weight ρ within a suitable range consistently improves the average reward and stabilizes convergence, while Fig. 12(c) demonstrates that performance exhibits a non-monotonic dependence on the observation range. *Discussions:* The experimental results reveal an intrinsic trade-off

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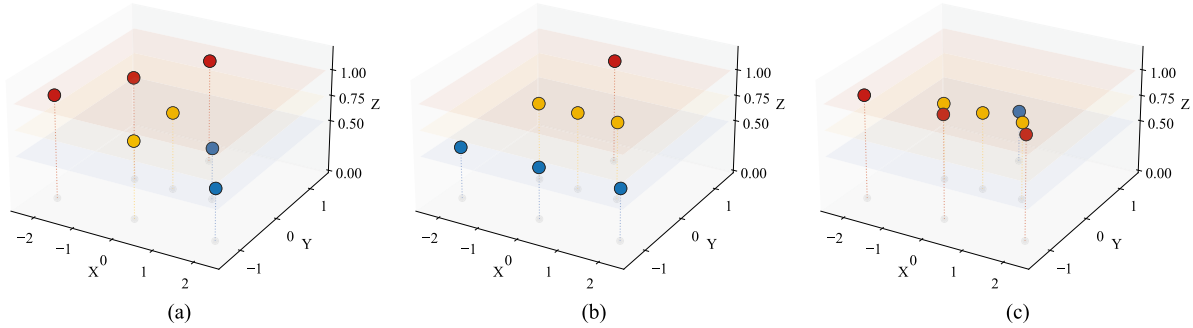


Fig. 11. An illustration of three representative 3D target formations. (a) Formation 1. (b) Formation 2. (c) Formation 3.

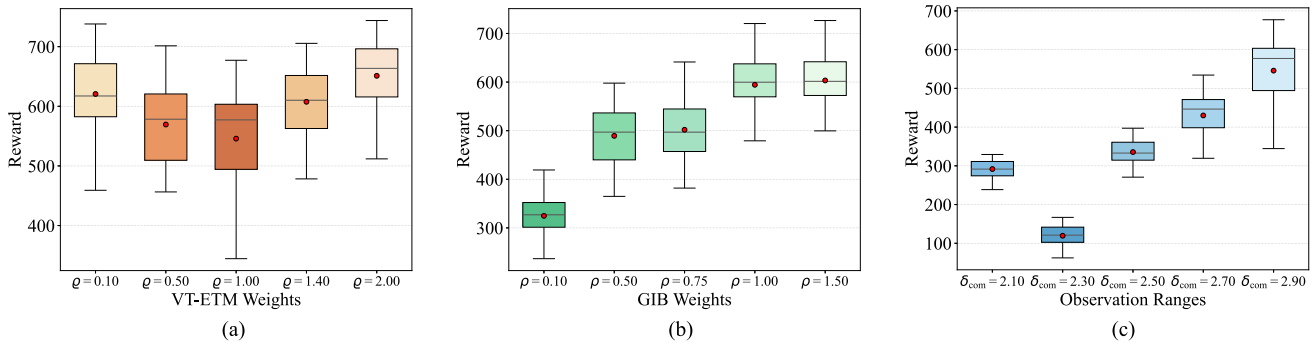


Fig. 12. Impact of hyperparameter configurations on reward performance. (a) Average reward with different VT-ETM weights ϱ. (b) Average reward with different GIB weights ρ. (c) Average reward with different maximum observation ranges δ_{com}.

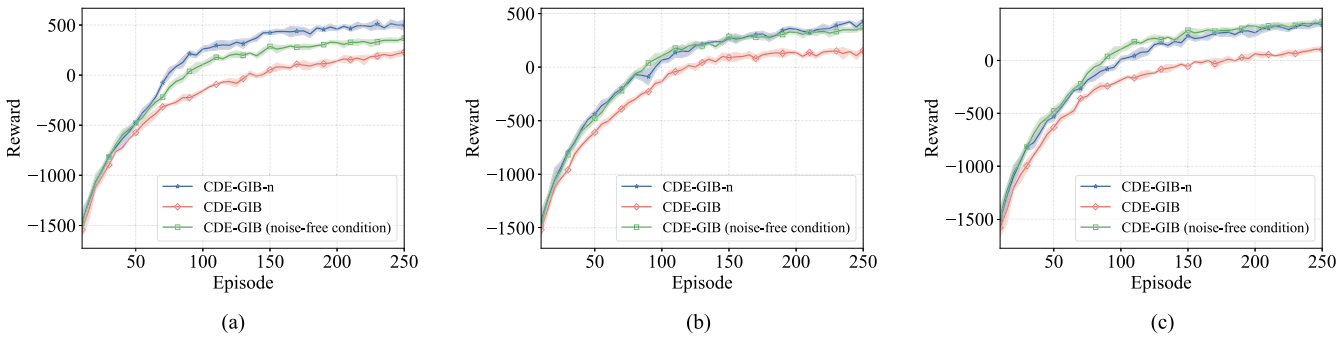


Fig. 13. Reward curves under different consensus noise levels σ_c. (a) $\sigma_c = 0.05$. (b) $\sigma_c = 0.10$. (c) $\sigma_c = 0.50$.

between communication intensity and decision-making efficacy in multi-agent systems. Adjusting the VT-ETM weights ϱ and observation ranges δ_{com} demonstrates that excessive message transmission, while initially improving coordination, introduces instability through possibly inconsistent local observations. Optimal performance necessitates balancing message quality and quantity, rather than simply pursuing raw data volume. Unlike the preceding approaches that regulate communication volume, GIB optimizes decision-making through content-aware compression while minimizing redundant inferred consensus.

G. Robustness of GIB Under Noise Perturbations

Experimental results in Fig. 13, obtained under a configuration of 7 agents and random seed 1, demonstrate the robustness and superiority of the proposed CDE-GIB-n algorithm under consensus noise perturbation. In particular, across all tested noise levels ($\sigma_c = 0.05$, 0.10, and 0.50), the version of CDE-GIB-n that incorporates both additive noise and noise-aware loss modeling consistently achieves the highest reward and a markedly faster convergence trend. In addition to consensus noise perturbations, we further evaluate the robustness of the proposed framework under channel noise characterized by different channel Signal-to-Noise Ratios (SNR = 0, 10, and 15 dB). Fig. 14 exhibits that CDE-GIB-n maintains stable convergence behavior and achieves consistently higher rewards compared with its counterparts across all tested SNR levels.

Discussions: The proposed CDE-GIB-n algorithm systematically validates its robustness against noise perturbations through explicitly modeling noise during GIB-driven feature

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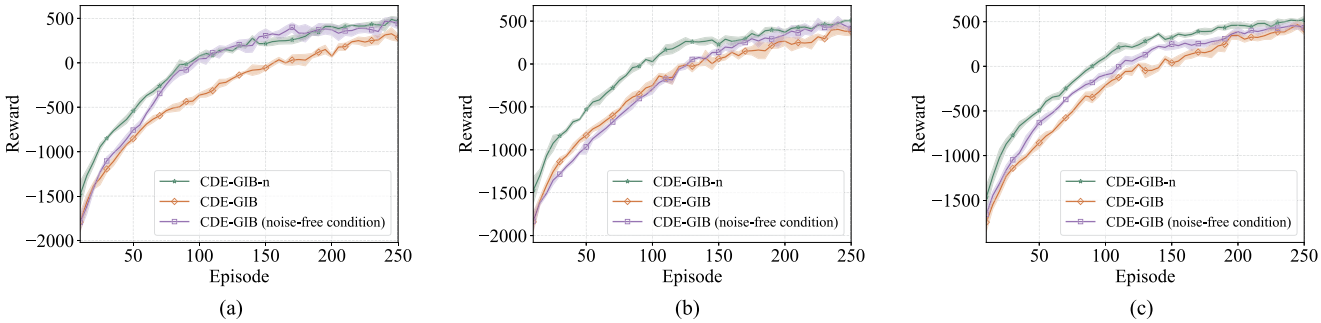


Fig. 14. Reward curves under different channel SNR levels. (a) SNR = 0 dB. (b) SNR = 10 dB. (c) SNR = 15 dB.

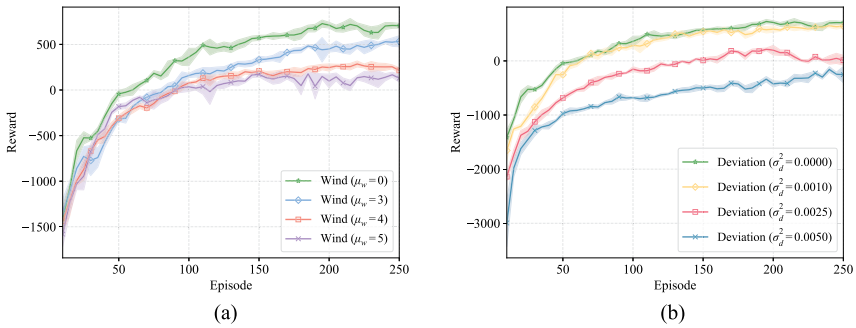


Fig. 15. Robustness evaluation under simulated real-world disturbances. (a) Performance under wind disturbances. (b) Performance under sensor measurement errors.

compression, which inherently suppresses stochastic signal fluctuations while prioritizing stable, high-dimensional semantic representations.

H. Validation Under Wind Disturbances and Sensor Inaccuracies

To address the practical applicability of our algorithm, we conduct comprehensive simulations replicating wind disturbances and sensor inaccuracies. As demonstrated in Fig. 15 with 7 agents with random seed 1, CDE-GIB maintains robust performance under these practical constraints. In particular, under Gaussian wind disturbances, the proposed method exhibits strong resilience and degrades gracefully as the disturbance intensity increases, rather than suffering catastrophic failure. A similar trend is observed under sensor measurement errors, where CDE-GIB continues to achieve effective learning convergence even as the noise level grows.

Discussions: These results collectively affirm CDE-GIB's practical viability. Beyond environmental and sensing noise, we also evaluate CDE-GIB under the Rician fading channel with communication disruptions such as random message drops and transmission delays. As shown in Fig. 16, the latency-only case slightly outperforms the ideal channel, suggesting that full, delay-free message exchange is not always optimal for coordination [19]. However, as packet loss increases, we observe a gradual performance decline, confirming that excessive message omission hinders cooperation.

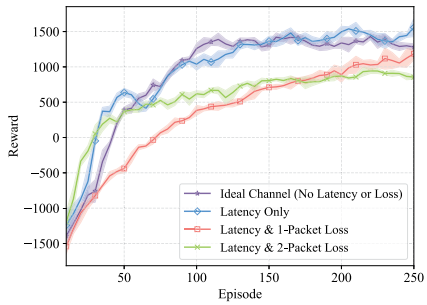


Fig. 16. Impact of latency and packet loss on system performance.

I. Ablation Experiments

In this subsection, we evaluate the effectiveness of the components incorporated in our proposed CDE-GIB under the Rician fading channel, with a configuration of 7 agents and random seed 1. As shown in Tables IV and V, the outcome clearly illustrates that each component significantly contributes to improving the information processing effectiveness despite channel impairments. Specifically, VT-ETM achieves substantial communication volume[4] reduction, with peak savings exceeding 25% at $\delta_{\text{com}} = 2.1$ m and an average reduction of 11.40% across all tested ranges. In terms of stability, VT-ETM demonstrates improved reliability in majority of practical operating conditions,

[4]Notably, the term "volume" in both Tables IV and V indicates the summation of dimensions of latent variables.

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TABLE IV
COMPARISON OF THE COMMUNICATION VOLUME (↓) WITH AND WITHOUT VT-ETM, UNDER DIFFERENT MAXIMUM OBSERVATION RANGES δ_{COM}

δ_{com} (m)	**2.1**	**2.3**	**2.5**	**2.7**	**2.9**
w. ETM	$\mathbf{666.41 \pm 12.86}$	$\mathbf{819.99 \pm 8.98}$	$\mathbf{700.45 \pm 19.52}$	$\mathbf{813.70 \pm 10.86}$	$\mathbf{763.72 \pm 8.67}$
w.o. ETM	898.85 ± 14.17	824.37 ± 5.82	744.84 ± 9.50	872.19 ± 14.47	930.45 ± 18.99

TABLE V
COMPARISON OF INFERRED CONSENSUS VOLUME (↓) WITH AND WITHOUT GIB, UNDER DIFFERENT MAXIMUM OBSERVATION RANGES δ_{COM}

δ_{com} (m)	**2.1**	**2.3**	**2.5**	**2.7**	**2.9**
w. GIB	$\mathbf{185.81 \pm 5.84}$	$\mathbf{190.95 \pm 1.94}$	$\mathbf{200.00 \pm 3.50}$	$\mathbf{188.03 \pm 3.49}$	$\mathbf{190.31 \pm 2.42}$
w.o. GIB	595.72 ± 12.48	634.10 ± 5.40	625.36 ± 5.05	601.88 ± 4.57	589.97 ± 4.70

achieving up to 54.34% variance reduction at $\delta_{\text{com}} = 2.9$ m, despite occasional variance increases at specific ranges. GIB delivers consistent compression benefits, achieving an average $3.19\times$ reduction in consensus inference volume. This high compression ratio remains stable as distance increases, highlighting GIB's robustness in maintaining efficient communication despite varying environmental conditions.

Discussions: VT-ETM benefits the removal of unsuitable collaborators, resulting in effectively reducing the frequency of event triggers and communication volume. Analogously, the gain from GIB is obtained due to efficiently fusing the communication graph and information flow for more concise message representations. Thus, both techniques can effectively enhance the information processing performance.

VI. CONCLUSION

In this work, we have proposed and validated CDE-GIB, a robust, event-triggered integrated communication and control framework with GIB optimization. To be specific, we have implemented a GIB module that jointly optimizes the communication graph and information flow in ConsMAC methodology. This module has effectively compressed consensus information into a compact and sufficient representation by learning an appropriate embedding that condenses messages derived from both communication graph and information flow, while closely approximating the target global observation label. Additionally, a VT-ETM has been employed to assess the information importance based on the fusion of historical data and current observations. Leveraging this temporal context has attenuated sensitivity to transient fluctuations and substantially reduced superfluous communication updates. This adaptable design has been further reinforced by its threshold variability that dynamically adjusts the triggering criteria to curb the accumulation of prediction errors over extended model rollouts. Moreover, we have reformulated a noise-resilient variant of the GIB to better combat the cumulative errors in the multi-agent interactions, which inherently suppresses stochastic signal variations while emphasizing stable, high-dimensional semantic representations. We have conducted extensive experiments to demonstrate the effectiveness, adaptability and robustness of our proposed method in communication-limited environments. In future work, we will further pursue more expressive distributions for the GIB, explore larger-scale formations under stricter communication constraints and deploy the approach on a more practical hardware platform.

REFERENCES

[1] X. HeD. Luo, Z. Chen, G. Yang, and Y. Liu, "Consensus based reinforcement learning method for differentiated formation control of UAVs," *IEEE Trans. Veh. Technol.*, vol. 74, no. 2, pp. 2429–2442, Feb. 2025.

[2] C. Yao, H. Tian, C. Wang, L. Song, J. Jing, and W. Ma, "Joint optimization of control and communication in autonomous UAV swarms: Challenges, potentials, and framework," *IEEE Wireless Commun.*, vol. 28, no. 5, pp. 28–35, Oct. 2021.

[3] A. Haydari and Y. Yilmaz, "Deep reinforcement learning for intelligent transportation systems: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 11–32, Jan. 2022.

[4] L. Ding, Q. Mao, and G. Yan, "Observer-based multiagent deep reinforcement learning: A fully distributed training scheme," *IEEE Trans. Ind. Inf.*, vol. 21, no. 2, pp. 1150–1159, Feb. 2025.

[5] A. Das et al., "TarMAC: Targeted multi-agent communication," in *Proc. Mach. Learn. Res.*, Jun. 2019, pp. 1538–1546.

[6] C. Guan et al., "Efficient multi-agent communication via self-supervised information aggregation," in *Proc. Adv. Neural Inf. Proces. Syst.*, Nov. 2022, pp. 1020–1033.

[7] Y. Xiang, S. Li, R. Li, Z. Zhao, and H. Zhang, "Decentralized consensus inference-based hierarchical reinforcement learning for multiconstrained UAV pursuit-evasion game," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 18229–18243, Oct. 2025.

[8] Y. Niu et al., "Multi-agent graph-attention communication and teaming," in *Proc. 20th Int. Conf. Auton. Agents Multiagent Syst.*, May 2021, pp. 964–973.

[9] M. Krouka, A. Elgabli, C. B. Issaid, and M. Bennis, "Communication-efficient split learning based on analog communication and over the air aggregation," in *Proc. IEEE Glob. Commun. Conf.*, Dec. 2021, pp. 1–6.

[10] T.-Y. Tung, S. Kobus, J. P. Roig, and D. Gündüz, "Effective communications: A joint learning and communication framework for multi-agent reinforcement learning over noisy channels," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2590–2603, Aug. 2021.

[11] N. Tishby et al., "The information bottleneck method," 2000, *arXiv:physics/0004057*.

[12] G. He, S. Cui, Y. Dai, and T. Jiang, "Learning task-oriented channel allocation for multi-agent communication," *IEEE Trans. Veh. Technol.*, vol. 71, no. 11, pp. 12016–12029, Nov. 2022.

[13] T. Wu et al., "Graph information bottleneck," in *Proc. Adv. Neural Inf. Proces. Syst.*, Dec. 2020, pp. 20437–20448.

[14] S. Ding, W. Du, L. Ding, J. Zhang, L. Guo, and B. An, "Robust multi-agent communication with graph information bottleneck optimization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3096–3107, May 2024.

[15] M. Lee, G. Yu, H. Dai, and G. Y. Li, "Graph neural networks meet wireless communications: Motivation, applications, and future directions," *IEEE Wireless Commun.*, vol. 29, no. 5, pp. 12–19, Oct. 2022.

[16] J. Xia et al., "GNN cleaner: Label cleaner for graph structured data," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 2, pp. 640–651, Feb. 2024.

[17] M. Zhang et al., "Robust heterogeneous graph neural networks against adversarial attacks," in *Proc. AAAI Conf. Artif. Intell., AAAI*, Jun. 2022, pp. 4363–4370.

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[18] D. Kim et al., "Learning to schedule communication in multi-agent reinforcement learning," in *Proc. Int. Conf. Learn. Representations*, May 2019, pp. 1–17.
[19] J. Jiang et al., "Learning attentional communication for multi-agent cooperation," in *Proc. Adv. Neural Inf. Proces. Syst.*, NIPS, Dec. 2018, pp. 7265–7275.
[20] G. Hu, Y. Zhu, D. Zhao, M. Zhao, and J. Hao, "Event-triggered communication network with limited-bandwidth constraint for multi-agent reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 3966–3978, Aug. 2023.
[21] Z. He, G. Wang, Y. Fan, and S. Qiao, "Fixed threshold event-triggered adaptive sliding mode trajectory tracking control of unmanned surface vehicles," in *Proc. 17th IEEE Conf. Ind. Electron. Appl.*, Dec. 2022, pp. 33–39.
[22] S. Li, Y. Wang, S. Guo, and C. Feng, "Task-oriented communication for graph data: A graph information bottleneck approach," *IEEE Trans. Cognit. Commun. Netw.*, vol. 11, no. 3, pp. 1723–1737, Jun. 2025.
[23] R. Lowe et al., "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Proc. Adv. Neural Inf. Proces. Syst.*, Dec. 2017, pp. 6382–6393.
[24] Y. Xiang, S. Li, R. Li, Z. Zhao, and H. Zhang, "Decentralized adaptive formation via consensus-oriented multi-agent communication," in *Proc. IEEE Int. Conf. Wireless Commun. Signal Process.*, Jul. 2023, pp. 449–454.
[25] C. Liu, L. Liu, and Z. Wu, "Intermittent event-triggered optimal control for second-order delayed multiagent systems with input constraints," *IEEE Trans. Syst. Man Cybern.: Syst.*, vol. 54, no. 5, pp. 2698–2710, May 2024.
[26] L. Zhong, Z. Chen, H. Cheng, and J. Li, "Lightweight federated graph learning for accelerating classification inference in UAV-assisted MEC systems," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21180–21190, Jun. 2024.
[27] Y. Wang et al., "Tom2c: Target-oriented multi-agent communication and cooperation with theory of mind," in *Proc. Int. Conf. Learn. Representations*, Apr. 2022, pp. 1–17.
[28] A. Singh et al., "Learning when to communicate at scale in multiagent cooperative and competitive tasks," in *Proc. Int. Conf. Learn. Representations*, May 2019, pp. 1–16.
[29] H. Mao et al., "Learning agent communication under limited bandwidth by message pruning," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2020, pp. 5142–5149.
[30] Y. Xie, S. Mou, and S. Sundaram, "Communication-efficient and resilient distributed Q-learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3351–3364, Mar. 2024.
[31] X. Wang, S. Baldi, X. Feng, C. Wu, H. Xie, and B. De Schutter, "A fixed-wing UAV formation algorithm based on vector field guidance," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 1, pp. 179–192, Jan. 2023.
[32] S. Baldi, D. Sun, G. Zhou, and D. Liu, "Adaptation to unknown leader velocity in vector-field UAV formation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 1, pp. 473–484, Feb. 2022.
[33] S. Gong, M. Wang, B. Gu, W. Zhang, D. T. Hoang, and D. Niyato, "Bayesian optimization enhanced deep reinforcement learning for trajectory planning and network formation in multi-UAV networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10933–10948, Aug. 2023.
[34] A. A. Khuwaja, Y. Chen, N. Zhao, M.-S. Alouini, and P. Dobbins, "A survey of channel modeling for UAV communications," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2804–2821, Fourthquarter 2018.
[35] B. Taşkesen et al., "Optimality of linear policies in distributionally robust linear quadratic control," 2025, *arXiv:2508.11858*.
[36] D. Tse et al., *Fundamentals of Wireless Communication*. Cambridge, U.K.: Cambridge Univ. Press, 2005.
[37] S. Triantafyllou et al., "Actual causality and responsibility attribution in decentralized partially observable markov decision processes," in *Proc. AAAI Conf. Artif. Intell.*, Aug. 2022, pp. 739–752.
[38] C. Yu et al., "The surprising effectiveness of PPO in cooperative multi-agent games," in *Proc. Adv. Neural Inf. Proces. Syst.*, Nov. 2022, pp. 24611–24624.
[39] V.-L. Heuthe et al., "Counterfactual rewards promote collective transport using individually controlled swarm microrobots," *Sci. Rob.*, vol. 9, no. 97, Dec. 2024, Art. no. eado5888.
[40] A. Vaswani, "Attention is all you need," in *Proc. Adv. Neural Inf. Proces. Syst.*, Dec. 2017, pp. 6000–6010.
[41] W. Chen, H. Du, and C.-C. Chen, "Stability and robustness analysis of finite-time consensus algorithm for second-order multiagent systems under sampled-data control," *IEEE Trans. Syst. Man Cybern.: Syst.*, vol. 53, no. 3, pp. 1445–1452, Mar. 2023.
[42] C. Ma et al., "Efficient and scalable reinforcement learning for large-scale network control," *Nature Mach. Intell.*, vol. 6, no. 9, pp. 1006–1020, Sep. 2024.
[43] B. Cheng, Z. Wu, and Z. Li, "Distributed edge-based event-triggered formation control," *IEEE Trans. Cybern.*, vol. 51, no. 3, pp. 1241–1252, Mar. 2021.
[44] A. A. Alemi et al., "Deep variational information bottleneck," in *Proc. Int. Conf. Learn. Representations*, Apr. 2017, pp. 1–19.
[45] K. Jiang et al., "Recovering from out-of-sample states via inverse dynamics in offline reinforcement learning," in *Proc. Adv. Neural Inf. Proces. Syst.*, Dec. 2023, pp. 38815–38826.
[46] C. Xie et al., "A statistical online inference approach in averaged stochastic approximation," in *Proc. Adv. Neural Inf. Proces. Syst.*, Nov. 2022, pp. 8998–9009.
[47] M. Seeger, "Gaussian processes for machine learning," *Int. J. Neural Syst.*, vol. 14, no. 02, pp. 69–106, Nov. 2004.
[48] R. Sutton and A. G. Barto et al., "Reinforcement learning: An introduction," *IEEE Trans. Neural Netw.*, vol. 9, no. 5, pp. 1054–1054, Sep. 1998.



Ziqiong Wang received the B.E. degree in communication engineering from Xidian University, Xi'an, China, in 2024. She is currently working toward the Ph.D. degree with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, China. Her research interests include communications in distributed learning and reinforcement learning.



Xiaoxue Yu received the B.E. degree in communication engineering from Xidian University, Xi'an, China. She is currently working toward the Ph.D. degree with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, China. Her research interests include communications in distributed learning and multiagent reinforcement learning.



Rongpeng Li (Senior Member, IEEE) received the B.E. degree from Xidian University, Xi'an, China, in 2010, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2015. From 2015 to 2016, he was a Research Engineer with the Wireless Communication Laboratory, Huawei Technologies Company Ltd., Shanghai, China. He was a Visiting Scholar with the Department of Computer Science and Technology, University of Cambridge, Cambridge, U.K., from February 2020 to August 2020. He is currently an Associate Professor with the College of Information Science and Electronic Engineering, Zhejiang University. His research focuses on networked intelligence for comprehensive efficiency (NICE).



Zhifeng Zhao received the B.E. degree in computer science, the M.E. degree in communication and information systems, and the Ph.D. degree in communication and information systems from the PLA University of Science and Technology, Nanjing, China, in 1996, 1999, and 2002, respectively. From 2002 to 2004, he was a Postdoctoral Researcher with Zhejiang University, Hangzhou, China, where his researches were focused on multimedia next-generation networks (NGNs) and softswitch technology for energy efficiency. He is currently a Chief Engineering Officer with the Zhejiang Lab, Hangzhou. His research interests include software defined networks, wireless network in 6 G, computing networks, and collective intelligence. He is the Symposium Co-Chair of ChinaCom 2009 and 2010. He is the Technical Program Committee, Co-Chair of the 10th IEEE International Symposium on Communication and Information Technology (2010).

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